Q1: What is Study of trace fossil is called? 2

The study of trace fossils, often called ichnology (from the Greek ichnos, a trace).

Q2: What is meant by Biogeography? 2

Biogeography is the study of the distribution of species and ecosystems in geographic space and through geological time.

Q3: Redundant species hypothesis? 3

According to the *redundant species hypothesis*, the removal of certain species from an ecosystem would have little or no effect on the functioning of that ecosystem. Here, the function of an ecosystem could refer to its overall productivity, its rate of nutrient cycling or its general self- sustainability.

Q4: What are characteristic features of Pterosaurs? 3

The pterosaurs were proficient flapping flyers, with a lightweight body, narrow hatchet-shaped skull and a long narrow wing supported on a spectacularly elongated fourth finger of the hand. Pterosaurs were covered with hair, and were almost certainly **endothermic**. Most pterosaurs fed on fishes caught in coastal seas, but others were insectivorous.

Q5: Functions of Geographic Information System.

Modern biogeography often employs the use of Geographic Information Systems (GIS), to understand the factors affecting organism distribution, and to predict future trends in organism distribution. Often mathematical models and GIS are employed to solve ecological problems that have a spatial aspect to them.

Q6: History of old world tropical mammal fauna of Africa 5

Africa contains the best known and most distinctive of the Old World tropical mammal fauna. Molecular studies of the interrelationships of the different orders of placentals suggest that the common ancestor of a group of African placentals entered the continent in the Late Cretaceous. This group is made up of the elephants, hyracoids (conies), aquatic sirenians (sea cows), elephant shrews, aardvarks, Cape golden moles, insectivorous tenrecs and extinct embrithopods; they have been placed together in a super order, Afrotheria, that is endemic to Africa.
The early horse *Hipparion*, which had evolved in North America, appeared in both Eurasia and Africa, while rhinoceros, hyenas and sabre- toothed cats dispersed from Africa to Eurasia.
The elephants originated in Africa in the Eocene, migrated into Eurasia in the Early Miocene and migrated back later in the Miocene. The ancestors of the mammoths and modern Asian elephant probably evolved in Africa 4 to 5 mya and migrated back into Eurasia 2 million years later.

Q7: What are three types of radiated amniotes 5

Amniotes radiated during the Late Carboniferous, giving rise to three main clades. These are distinguished by the pattern of openings in the side of the skull, especially the **temporal openings** behind the eye socket. These three skull patterns diagnose the key clades among amniotes which are

i. The anapsid ("no arch") is the primitive state skull pattern, since there are no temporal openings. The Anapsida include various early forms such as *Hylonomus*, as well as some Permian and Triassic reptiles, and the turtles. The two other skull patterns seen in amniotes are

ii. The synapsid ("same arch"), where there is a lower temporal opening. The Synapsida include the "mammal-like reptiles" and the mammals.

iii. The diapsid ("two arch") pattern, where there are two temporal openings. The Diapsida includes a number of early groups, as well as the lizards and snakes and the crocodiles, pterosaurs, dinosaurs and birds.

Q8: Work of Leon Croizat in field of Biogeography.

He amassed a vast array of distributional data, representing each biogeographical pattern as a line, or track, connecting its known areas of distribution. Croizat studied the distribution patterns of many unrelated taxa, and for each he drew lines or 'tracks' on the map linking the areas in which they are found. In many cases, these lines were similar enough in position to be combined as 'generalized tracks', shown here. He found that the tracks of many taxa, belonging to a wide variety of organisms, could be combined to form a generalized track that connected different regions of the world

Q9: Write a note on geographical history of New Caledonia

New Caledonia

Geological studies show that New Caledonia is the smallest (about 1600 km2) surviving fragment of Gondwana, having rifted away from Australia in the Late Cretaceous 80–65 mya and reaching its present position 1500 km east of it 50 mya. It is a hotspot for diversity: 77% of its flowering plant species are endemic to the island, as are 98% of the species of the flowering plant family Sapotaceae.

In view of the large amount of research that shows the frequency of long- distance dispersal in such scenarios, most biogeographers today believe that this was the mechanism of arrival of the ancestors of the **New Caledonian taxa.**

Geological studies show that the island was submerged for long periods in the **Paleocene– Eocene**, so that the island was not available for colonization until about 37 mya. The New Caledonian species must therefore have diverged from their ancestors after this date.

Panbiogeographers, however, **reject long- distance dispersal**. They therefore have to theorize that the Cretaceous ancestors of the present biota were widespread across the Pacific islands, including on supposedly formerly exposed large plateaux such as the Ontong Java plateau.

Biogeographers in Sweden and France have investigated the history of the New Caledonian species of the angiosperm family Sapotaceae. They found that these resulted from nine separate colonizations of the island between 33 and 4.2 mya, dates that are about 40 million years after the rifting of New Caledonia from Australia. New Guinea and Australia were the most important sources of the arrivals, those from Pacific islands arriving only more recently, 27–24 mya. These results are clearly totally incompatible with the panbiogeographers' theories.

Q10: What are two key defining characters of Jawless fishes? 2

Two key defining characters: The head and neural crest tissues.

The neural crest appears in the early embryo as a strip of cells lying just below the outer skin, the ectoderm, of the embryo, above the line of where the backbone will develop.

Q11: Oldest clear evidence for bipedalism

The oldest clear evidence for bipedalism was the find of some human tracks in volcanic ash from Tanzania, dated at 3.75 Ma. The oldest substantial skeletons, of *Praeanthropus afarensis*, come from rocks dated at about 3.2 Ma and also show clear anatomic evidence for advanced bipedalism, but still an ape-sized brain.

Q12: What is Kleptoparasitisim with example? 5

Kleptoparasitism (literally, parasitism by theft) is a form of feeding in which one animal takes prey or other food from another that has caught, collected, or otherwise prepared the food, including stored food (as in the case of cuckoo bees, which lay their eggs on the pollen masses made by other bees; food resources could also be in the form of hosts of parasitic or parasitoid wasps). The term is also used to describe the stealing of nest material or other inanimate objects from one animal by another.

One very widespread duck in both continents is the gadwall. It feeds mainly on submerged vegetation, which is most abundant in nutrient- rich (eutrophic) waters. It cannot dive, so it feeds by upending, and this generally restricts it to the shallow parts of a lake, where it may encounter some competition from other water birds, such as the mallard. Coots (both Eurasian coot and the American coot dive for their food and bring vegetable matter to the surface from greater depths. The coots are messy eaters, and it is not difficult for gadwall to move in and collect some of the loot. This behavior is called **kleptoparasitism**, and it is an effective way of widening the niche of the gadwall.

Q13: Biogeography of South America 10

South America

The history of the biota of South America has been dominated by the effects of the great earth-engine of plate tectonics. Some of these effects were caused by the westward drift of the continent, which led to mountain building and consequent climatic changes, but most were the result of changes in its relationship with North America. As noted in the description of the origin of the Caribbean islands in the 'The West Indies' section, there had been islands in that region ever since the Americas started to move westward in the Early Cretaceous. This may have been the route by which some North American dinosaurs dispersed southward in the Mid-Cretaceous, accompanied by some early types of marsupial and placental mammal. But this connection has always been tenuous, sometimes permitting passage and sometimes becoming broken. This led to cycles of immigration, isolation and evolution, and final new immigration and extinction.

The Early Cenozoic:

The mammal fauna of South America today is characterized by a few marsupials (opossums) and a diversity of edentate placental mammals known as the Xenarthra (sloths, armadillos and South American anteaters) that are hardly known elsewhere. However, in the Late Cretaceous, South America had a mammal fauna of strange marsupials and of ungulate placentals quite unlike anything known elsewhere in the world. Alfred Wallace, one of the founders of zoogeography noted that the range boundaries of Amazonian birds seemed to coincide with the pattern of the rivers. It is fascinating that recent research shows that closely related taxa of some birds are to be found on either side of rivers throughout Amazonia.

Q14: Describe the work of James Dana, and Edward Forbes in the Biogeography of Ocean 5

American scientist **James Dana**, who later became an eminent geologist. His brief paper, published in 1853, divided the surface waters of the globe into several different zones based on their mean minimum temperature.

Three years later 1856, the British zoologist **Edward Forbes** published the first comprehensive work, recognizing five depth zones and 25 faunal provinces along the coasts of the continents. He was the first to recognize the enormous Indo- Pacific faunal region.

Q15: What is meant by Niche? 3

The demands that an organism places on its environment in terms of physical and chemical conditions, space and food supply help to define what ecologists call its **niche**. But the concept of the niche goes beyond the basic physics and chemistry of its habitat and covers all aspects of how the organism makes a living. It includes the food an animal requires, but also encompasses the way in which it acquires that food. The kestrel is a bird that hunts small mammals by day, whereas an owl performs a similar activity by night. Swallows catch aerial insects by day, and bats have the same feeding strategy at night.

Q16: What was the brain capacity of Homo habilis? 3

The first species, *H. habilis*, lived in Africa from 2.4 to 1.5 Ma, and had a brain capacity of 630–700 cm3 in a body only 1.3 m tall. *H. habilis* may have used tools.

Q17: Name the most famous fossils. 3

One of the most famous fossils is *Archaeopteryx*, the oldest known bird. The first specimen was found in Upper Jurassic sediments in southern Germany in 1861, and was hailed as the ideal "missing link" or proof of evolution in action. Here was an animal with a beak, wings and feathers, so it was clearly a bird, but it still had a reptilian bony tail, claws on the hand, and teeth. Since 1861, nine more skeletons have come to light, the last two in 1992 and 2005.

Q18: Name the old amniotes and where it was found?

The oldest-known amniote, *Hylonomus* from the mid-Carboniferous of Canada is known only from its skeleton; no amniotic eggs are known from the Carboniferous.

Q19: What is the Biotic Replacement, give example?

Biotic Replacements are an obvious feature of the history of life. These are times when one group of plants or animals replaces another. The replacement of brachiopods by bivalves is a famous example. This had always been seen as a progressive process: the common view is that brachiopods are less adaptable than bivalves, and they clearly succumbed to long-term competition, perhaps lasting for tens or hundreds of millions of years.

Q20: Three amniotic layers in embryo and their functions.

- Enclose the embryo (the amnion)
- Collect waste (the allantois)
- line the eggshell (the chorion)
- Yolk, a yellow material rich in protein

Q21: Trace fossils excrement names, how they are different to other? 3

***Fecal pellets and fecal strings*:** small pellets, usually less than 10 mm in length, or strings of excrement.
***Coprolites*:** discrete fecal masses, usually more than 10 mm in length, and usually the product of vertebrates.

Q22: Describe the Island biogeography of New Candolle 3

Another early contributor was Candolle, who pointed out that the age, climate and degree of isolation of an island, and whether or not it was volcanic, would also affect the diversity of its flora. Nevertheless, the sheer variety and volume of the works on island biogeography published by Alfred Wallace mark him as the real founder of studies on this subject.

Q23: The work of Albert Gunther, and Arbold Ortman in Biogeography 3

In 1880, the British zoologist **Albert Gunther** published a book on fishes in which he recognized 10 different regions in the distribution of shore fishes.
The German **Arnold Ortmann** published a similar work based on the distribution of crustaceans such as crabs and lobsters.

Q24: When modern birds appear?

Modern groups of birds appeared in the latest Cretaceous and Early Tertiary, including flightless ratites and ancestors of water birds, penguins and birds of prey. The perching birds or songbirds, consisting of 5000 species today, radiated in the Miocene

Q25: Erythrosachus

Small and large meat eaters such as *Erythrosuchus* appeared, one of the first of the archosaurs, a group that was later to include the dinosaurs, pterosaurs, crocodilians and birds.

Q26: Traces on bedding planes

- **Tracks**: sets of discrete footprints, usually formed by arthropods or vertebrates
- **Trails:** continuous traces, usually formed by the whole body, either trave

ZOO503 MCQs Final and Midterm Spring 2018

Final Term MCQs

Q1: The Spirallians are morphologically diversed clade of protostome animals including all of

the following except

Cnidarians

Q2: The most advance mollusks are

Cephalopods

Q3: Amniotes radiated in

Late carboniferous

Q4: In early crocodilian were most diverse

Jurassic and Cretaceous

Q5: Oldest known bird found in

Upper Jurassic sediments

Q6: The oldest human was found is approximately lived ---------- ago

6Mya

Q7: German paleontologist ---------- established a classification of trace fossil based on behavior

Adolf Seilacher

Q8: ---------- are the observation of the appearance and disappearance of species

Patterns

Q9: Which Scientist translated Cuvier's idea into English in 1813

Robert Jameson

Q10: ---------- argues that all areas connected by one of these tracks had originally formed a single, continuous are that was inhabited by group concerned

Panbiogeography

Q11: Briggs used pathogen of endamicity of coastal faunas to identify location where there appears to be a zone of unusually rapid faunal change and then used this to distinguish ----------

----------zoogeographical regions

Q12: The sum of all niche requirements under ideal conditions when the species is given unimpeded access to resources is called

Fundamental Niche

Q13: ---------- noted that the range boundaries of Amazonia birds seemed to coincide with the patterns of river

Alfred Wallace

Q14: Geological study of new Caledonia show that the Island was submerged for long period in the Paleocene --Eocene so that Island was not available for colonization until about

37 mya

Q15: The ancestors of mammoths and modern Asian elephant probably evolved in

Africa

Q16: During the early Devonian 380mya the fossil record suggest that all the early amphibians and reptile groups evolved in the continents called

Euramerica

Q17: All of the following part of Gondwana Except

North America

Q18: Many studies have demonstrated that the richness of the species particularly plant increases with ----------

Altitude

Q19: Evolution a process by which new species are generated it essentially involve

Genetic Variation and Subsequent selection

Q20: The trade winds found in the both northern and southern hemisphere meet in the region of equators this is known as

Intertropical Convergence zone (ITCZ)

Q21: ---------- is a term sometimes used to describe waves of biological invasions that has

occurred in the wake of human invasions by either accidental, transport or deliberate introduction

Ecological Imperialism

ZOO503 MCQs Final and Midterm Spring 2018

Q22: When two species are attempting to tap a resource in the same way and when that resource in short supply the two are said to be in

Competition

Q23: The theory of Island biogeography published in

1967

Q24: All of the following are included in ecological biogeography except

Altitude

Q25: In terms of diversification of life during the early Cambrian ---------- was an important part of an ecosystem

Predation

Q26: The focus of human evolution was in

Africa

Q27: The reptile ancestors of the mammals are

Basal Synapsid

Q28: Chondricthyans is fish

Cartilaginous

Q29: ---------- is a term sometimes used to describe the wave of biological invasion that has occurred

Ecological imperialism

Q30: Biogeography is an integrative field of inquiry that unites concept and information from solving branches of science except

Any option except these (Ecology, Evolutionary Biology, Geology, Physical Geography)

Q31: ---------- in on origin of species where devoted to geographical distributions

Two Chapters

Q32: The oldest substantial skeleton of praeanthropus afarensis lived approximately ----------

ago

3.2 Mya

ZOO503 MCQs Final and Midterm Spring 2018

Q33: Certain species play key roles in the ecosystem, and when they are lost there is a sudden

drop in the capacity of the ecosystem to function…..

Rivert hypthosis

Q34: First fossil evidence of bipedalism…….

Australopithecus afarensis

Q35: What is not a part of pangea?

Any Option Except these (north China, south China, Tibet, Indochina and South-East Asia)

Q36: Early vertebrate skeleton was made of?

Cartilaginous

Q37: Amphibians and reptiles evolved in …..

Triassic

Q38: Crocodile radiate in…..

Jurassic

Q39: Lucy fossil is old myb?

3.2 Million years ago

Q40: Lagerstatten is

Soft bodied Fossil

Q41: The jawless fish Ostracoderns generally armored, had their heyday in.....

Devonian

Midterm MCQs

Q1: Paleontology Originate from Greek word “Ontos” which means

Creature

Q2: Archaeopetyx is true missing link between

Dinosaurs and Birds

Q3: Trilobites dominated assemblages in England and Wales area were described by

William Smith

ZOO503 MCQs Final and Midterm Spring 2018

Q4: Fossils in Fold belts are, except

In Excellent Shape

Q5: Where an organism live matters in which of the filters is preservation of the complete fossil record

Ecological Filters

Q6: In Climate change in life history of earth has oscillated at least ---------- between icehouse and greenhouse conditions

5 times

Q7: Scala Naturae is ----------

The Ladder of life

Q8: When local ecological conditions affect the form of an organism during its life time and consequently fossil is called

Ecophenotypic Effects

Q9: Rohde and Muller (2005) argued that there is ---------- periodicity in mass extinctions

62 myr

Q10: Modification of the Oparin ---------- Haldane biochemical model is

Hydrothermal Model

Q11: ----------Biomarker indicates that eukaryotes were present around at least 2.7 Ga

Lipid

Q12: Chromistans Includes all of the following, except

Protozoa

Q13: The Parazoan body plan is seen in

Sponges

Q14: Fossils Lagerstattan are assemblages of

Preserved soft bodied

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Gastropods

Q18: The acme of the group Gastropoda with the radiation of the siphonal carnivorous neogastropods was in

Cenozoic

Q19: Chelicerates are diversed and heterogeneous which includes

All of the above (Mites, Scorpions, Spiders)

Q20: Water Vascular system is present in

Echinoderms

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In 1880, the British zoologist Albert Gunther published a book on fishes in which he recognized 10 different regions in the distribution of shore fishes.

The German Arnold Ortmann published a similar work based on the distribution of crustaceans such as crabs and lobsters.

Q24: When modern birds appear?

Modern groups of birds appeared in the latest Cretaceous and Early Tertiary, including flightless ratites and ancestors of water birds, penguins and birds of prey. The perching birds or songbirds, consisting of 5000 species today, radiated in the Miocene

Q25: Erythrosachus

Small and large meat eaters such as Erythrosuchus appeared, one of the first of the archosaurs, a group that was later to include the dinosaurs, pterosaurs, crocodilians and birds.

Q26: Traces on bedding planes

• Tracks: sets of discrete footprints, usually formed by arthropods or vertebrates

• Trails: continuous traces, usually formed by the whole body, either trave

The Spirallians are morphologically diversed clade of protostome animals including all of the following except……..**Cnidarians**

Early life was mainly affected by …..…..**predation**

Amphibians and reptiles evolved in …..Triassic

Lucy fossil old myb…. **3.2 million years**

Reptilian ancestors of mammals had… **synapsid skull**

The most advance mollusks are……..**Cephalopods**

Amniotes radiated in……..**Late carboniferous**

In early corocodilian were most diverse……..**Jurassic and Cretaceous**

Crocodile radiate in…..**jurassic**

First fossil evidence of bipedalism…….***Australopithecus afarensis***

Crocodile more diverse… **Jurassic and cretaceous**

Oldest known bird found in…….**Upper Jurassic sediments**

The oldest human was found is approximately lived _____ago……**.6Mya**

German paleontologist ---------- established a classification of trace fossil based on behavior ……..**Adolf Seilacher**

____are the observation of the appearance and disappearance of species……..**Patterns**

Which Scientist translated Cuvier's idea into English in 1813……**Robert Jameson**

_____ argues that all areas connected by one of these tracks had originally formed a single, continuous are that was inhabited by group concerned ……**Panbiogeography**

Briggs used pathogen of endamicity of coastal faunas to identify location where there appears to be a zone of unusually rapid faunal change and then used this to distinguish ----------**-zoogeographical regions**

Brigg distinguish _____ zoogeographical regions…….**23**

The sum of all niche requirements under ideal conditions when the species is given unimpeded access to resources is called……..**Fundamental Niche**

____noted that the range boundaries of Amazonia birds seemed to coincide with the patterns of river……**Alfred Wallace**

Geological study of New Caledonia show that the Island was submerged for long period in the Paleocene_____Eocene so that Island was not available for colonization until about……**37 mya**

The ancestors of mammoths and modern Asian elephant probably evolved in……**Africa**

During the early Devonian 380mya the fossil record suggest that all the early amphibians and reptile groups evolved in the continents called……..**Euramerica**

All of the following part of Gondwana Except……..**North America**

Many studies have demonstrated that the richness of the species particularly plant increases with ----------**Altitude**

Evolution a process by which new species are generated it essentially involve……**Genetic Variation and Subsequent selection**

The trade winds found in the both northern and southern hemisphere meet in the region of equators this is known as……..**Intertropical Convergence zone (ITCZ)**

_____ is a term sometimes used to describe waves of biological invasions that has occurred in the wake of human invasions by either accidental, transport or deliberate introduction ……**Ecological Imperialism**

When two species are attempting to tap a resource in the same way and when that resource in short supply the two are said to be in…….**Competition**

The theory of Island biogeography published in…….**.1967**

All of the following are included in ecological biogeography except……**Altitude**

In terms of diversification of life during the early Cambrian _____ was an important part of an ecosystem……**Predation**

The focus of human evolution was in…….**Africa**

The reptile ancestors of the mammals are……..s**ynapsid**

Chondricthyans is fish……**Cartilaginous**

_____is a term sometimes used to describe the wave of biological invasion that has occurred……..**Ecological imperialism**

Biogeography is an integrative field of inquiry that unites concept and information from solving branches of science except……..Any option except these (Ecology, Evolutionary Biology, Geology, Physical Geography).

Biology of ocean……marine biology

---------- in on origin of species where devoted to geographical distributions…….**Two Chapters**

The oldest substantial skeleton of praeanthropus afarensis lived approximately ---------- ago…….3**.2 Mya**

Early vertebrate skeletion…… **cartilage**

Certain species play key roles in the ecosystem, and when they are lost there is a sudden drop in the capacity of the ecosystem to function…..**Rivert hypthosis**

Q1: Paleontology Originate from Greek word “Ontos” which means……**Creature**

Q. Today Extenction**….. 6th**

45. Q2: Archaeopetyx is true missing link between…….**Dinosaurs and Birds**
46. Trilobites dominated assemblages in England and Wales area were described by…….**William Smith**
47. Fossils in Fold belts are, except…….**In Excellent Shape**
48. Where an organism live matters in which of the filters is preservation of the complete fossil record……**Ecological Filters**
49. In Climate change in life history of earth has oscillated at least ---------- between icehouse and greenhouse conditions……..**5 times**
50. Scala Naturae is **----------The Ladder of life**
51. When local ecological conditions affect the form of an organism during its life time and consequently fossil is called ……..**Ecophenotypic Effects**
52. Rohde and Muller (2005) argued that there is__ periodicity in mass extinctions …….**62 myr**
53. Modification of the Oparin_____Haldane biochemical model is……**Hydrothermal Model**
54. ___Biomarker indicates that eukaryotes were present around at least 2.7 Ga…….**Lipid**
55. Chromistans Includes all of the following, except……..**Protozoa**
56. The Parazoan body plan is seen in…….**Sponges**
57. Fossils Lagerstattan are assemblages of…….**inadequate Preserved soft bodied**
58. The Spirallians are morphologically diversed clade of protostome animals including all of the following except……**Cinideria**
59. The most advanced mollusks are ……..**Cephalopods**
60. The “Belly footed” Mollusks are……**Gastropods**
61. The acme of the group Gastropoda with the radiation of the siphonal carnivorous neogastropods was in …….**Cenozoic**
62. Chelicerates are diversed and heterogeneous which includes….**All of the above (Mites, Scorpions, Spiders)**
63. Water Vascular system is present in……..**Echinoderms**